

8 (short)

- (a) If the probability that a family has n children is $P_n = \alpha p^n$ with $n \geq 1$ then show that

$$P_0 = 1 - \frac{\alpha p}{1 - p}$$

gives the probability that a family is childless. [5]

- (b) Show that

$$\frac{1}{(1-x)^{n+1}} = \binom{n}{n} + \binom{n+1}{n}x + \binom{n+2}{n}x^2 + \dots$$

using the binomial expansion. [5]

9 (short) The function $f(x, y)$ is given by

$$f(x, y) = 5x^3 - 2y^3 - 6x^2y + 54y.$$

Find and classify its stationary points. [10]

10 (long) An even function $f(t)$ is periodic with a period $T = 4$ and $f(t) = 4 - t^2$ for $0 \leq t \leq 2$.

- (a) Sketch $f(t)$ in the range $-4 \leq t \leq 4$. [5]

- (b) Show that

$$f(t) = \frac{8}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos\left(\frac{n\pi t}{2}\right)$$

is a Fourier series representation of $f(t)$. [15]

- (c) Deduce that

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

using the expression in (b). [10]